

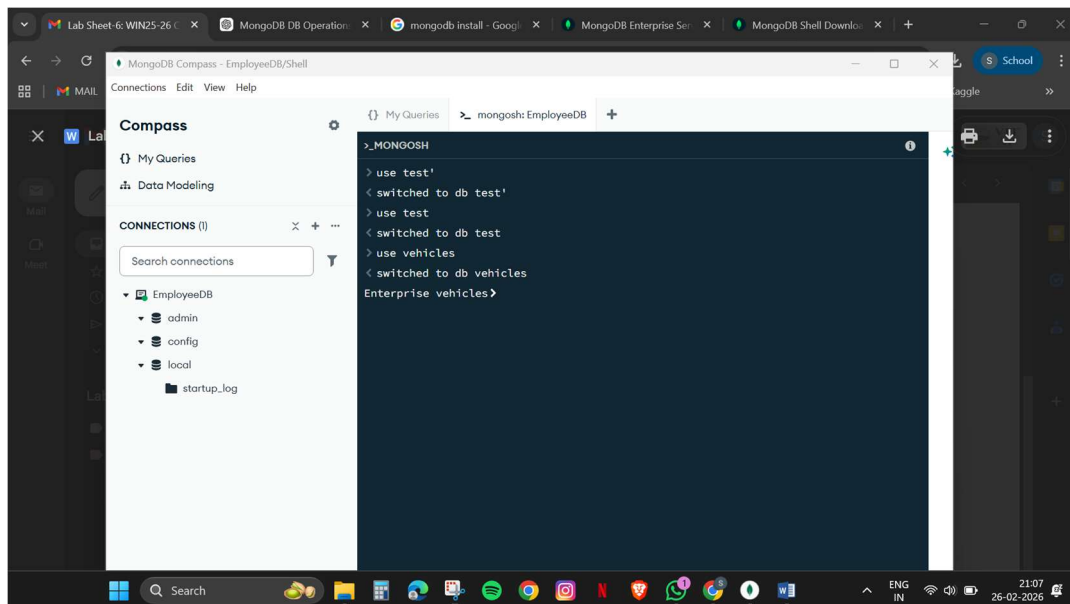
Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech
Faculty Name: Prof. S.Gopikrishnan
Student name: S.N. Swetha Shree

Date: 26.02.2026
School: SCOPE
Reg. no.: 23BCE8922

1. Use MongoDB to implement the following DB operations

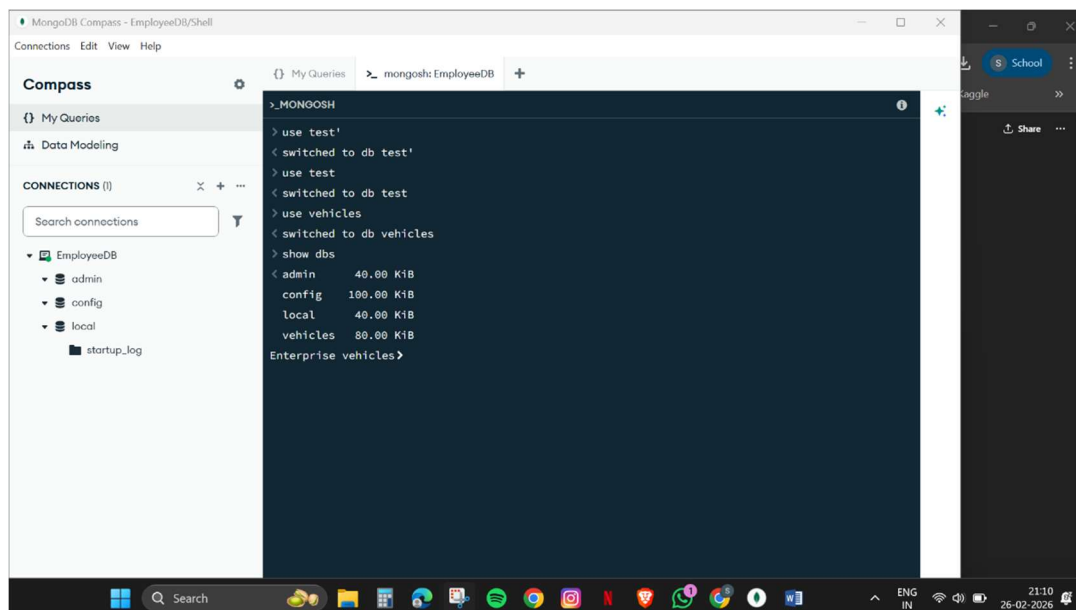
1. Create a database called 'vehicles' and *write* a MongoDB query to select database as "vehicles".



The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'Connections' panel with a tree view of the 'EmployeeDB' database, including collections 'admin', 'config', and 'local'. The main window shows the MongoDB Shell with the following commands and output:

```
>_MONGOSH
> use test'
< switched to db test'
> use test
< switched to db test
> use vehicles
< switched to db vehicles
Enterprise vehicles>
```

2. Write a MongoDB query to display all the databases.



The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'Connections' panel with a tree view of the 'EmployeeDB' database, including collections 'admin', 'config', and 'local'. The main window shows the MongoDB Shell with the following commands and output:

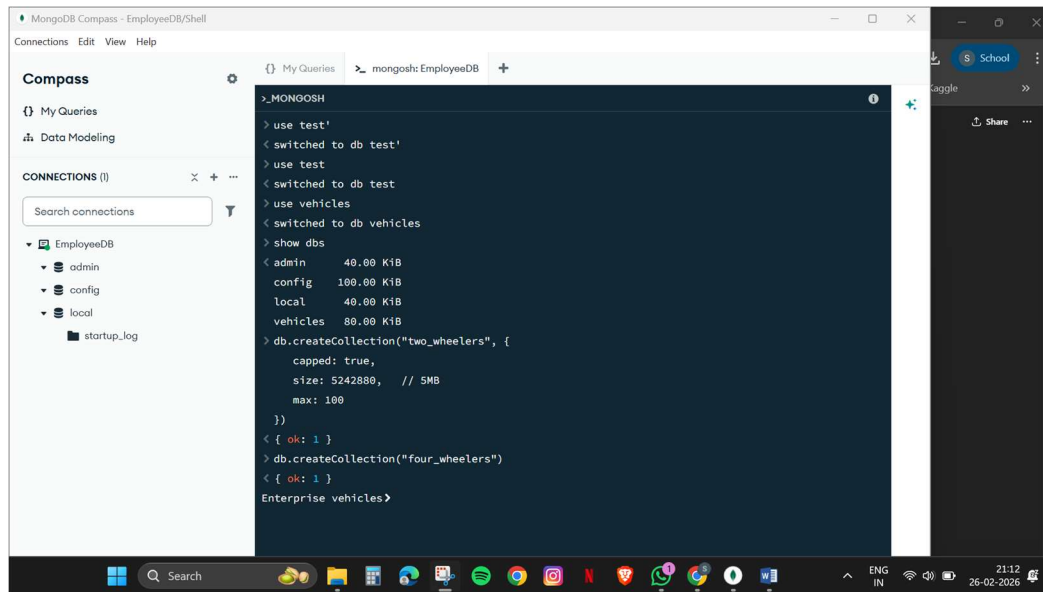
```
>_MONGOSH
> use test'
< switched to db test'
> use test
< switched to db test
> use vehicles
< switched to db vehicles
> show dbs
< admin    40.00 KiB
  config  100.00 KiB
  local   40.00 KiB
  vehicles 80.00 KiB
Enterprise vehicles>
```

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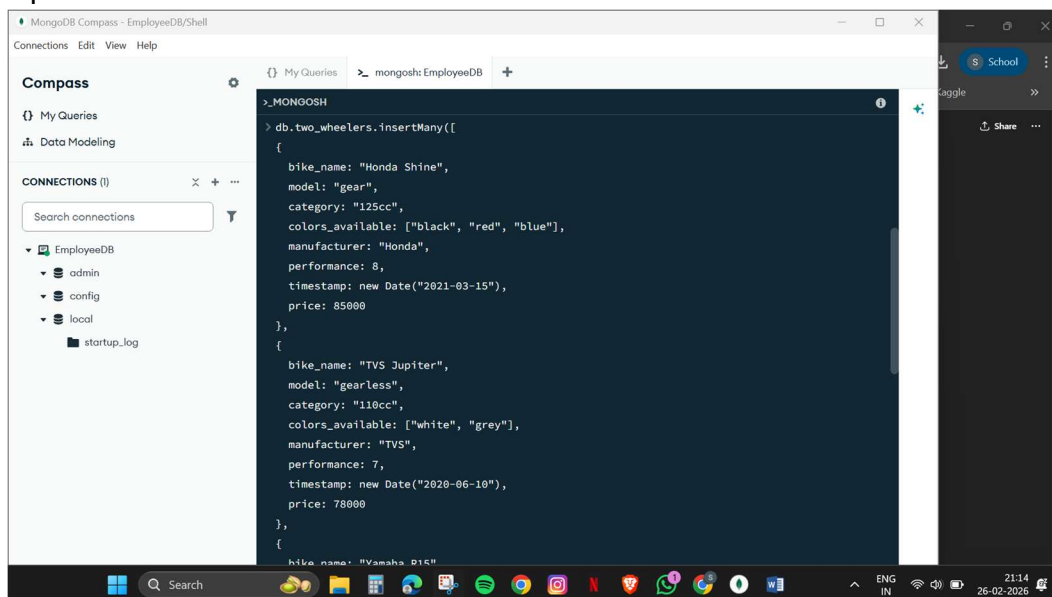
3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.



The screenshot shows the MongoDB Compass interface with the 'EmployeeDB' database selected. The 'MongoShell' tab is active, displaying the following commands and their outputs:

```
> use test
switched to db test
> use test
switched to db test
> use vehicles
switched to db vehicles
> show dbs
admin      48.00 KiB
config    100.00 KiB
local      48.00 KiB
vehicles   80.00 KiB
> db.createCollection("two_wheelers", {
  capped: true,
  size: 5242880, // 5MB
  max: 100
})
{ ok: 1 }
> db.createCollection("four_wheelers")
{ ok: 1 }
```

4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.



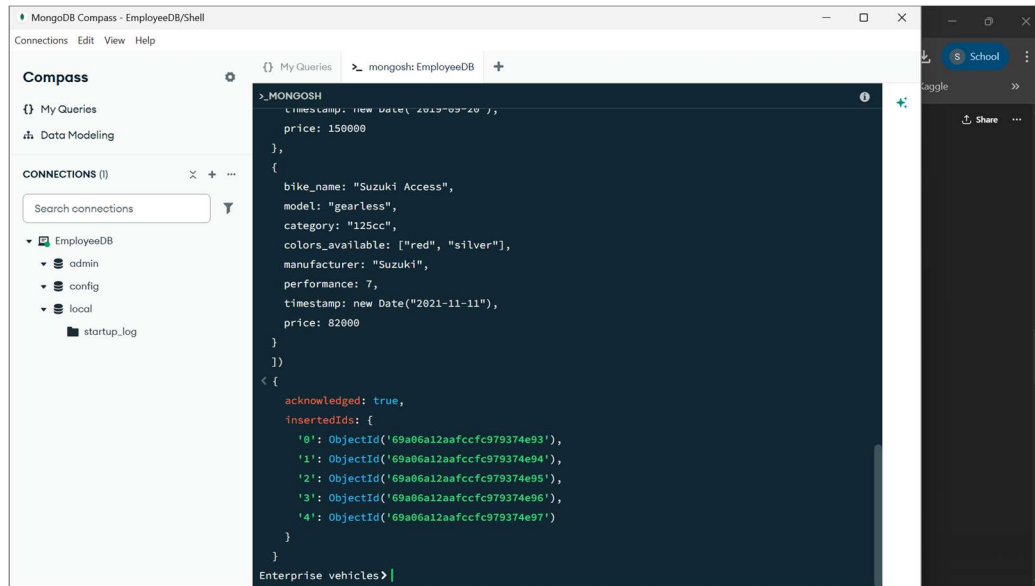
The screenshot shows the MongoDB Compass interface with the 'EmployeeDB' database selected. The 'MongoShell' tab is active, displaying the following command and its output:

```
> db.two_wheelers.insertMany([
  {
    bike_name: "Honda Shine",
    model: "gear",
    category: "125cc",
    colors_available: ["black", "red", "blue"],
    manufacturer: "Honda",
    performance: 8,
    timestamp: new Date("2021-03-15"),
    price: 85000
  },
  {
    bike_name: "TVS Jupiter",
    model: "gearless",
    category: "110cc",
    colors_available: ["white", "grey"],
    manufacturer: "TVS",
    performance: 7,
    timestamp: new Date("2020-06-10"),
    price: 78000
  },
  {
    bike_name: "Yamaha D15"
  }
])
```

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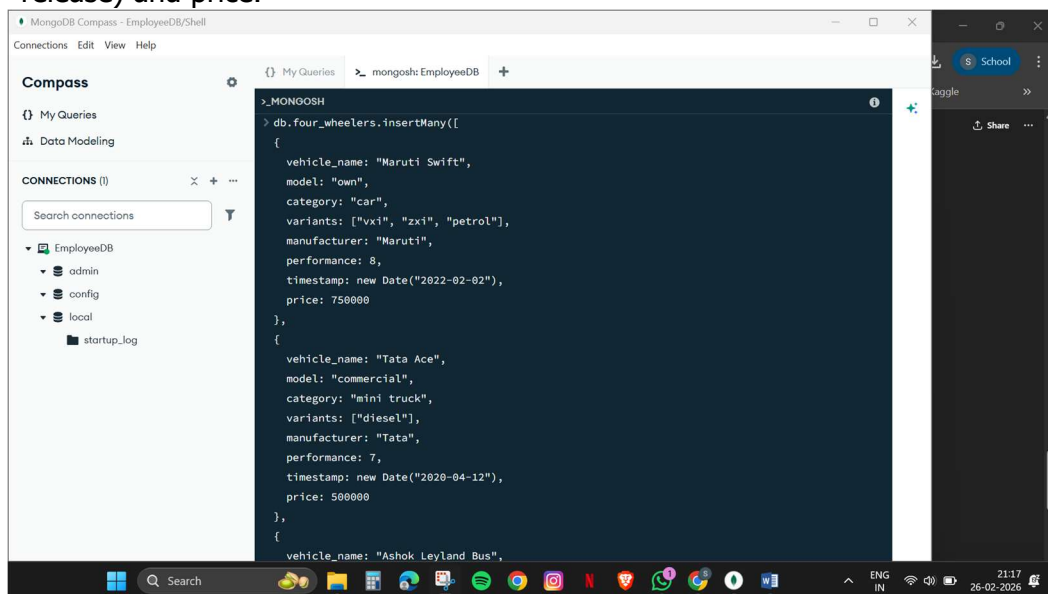
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The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists 'EmployeeDB' with sub-items 'admin', 'config', 'local', and 'startup_log'. The main window displays the 'My Queries' tab with a MongoDB shell session. The session shows a successful insertMany operation into the 'vehicles' collection. The output includes the 'acknowledged' status (true) and a list of five 'insertedIds' (ObjectIds). The document structure for the first vehicle is visible: { 'price': 150000, 'bike_name': 'Suzuki Access', 'model': 'gearless', 'category': '125cc', 'colors_available': ['red', 'silver'], 'manufacturer': 'Suzuki', 'performance': 7, 'timestamp': new Date('2021-11-11'), 'price': 82000 }.

```
> MONGOSH
timesComp: new Date('2023-03-20'),
price: 150000
},
{
  bike_name: "Suzuki Access",
  model: "gearless",
  category: "125cc",
  colors_available: ["red", "silver"],
  manufacturer: "Suzuki",
  performance: 7,
  timestamp: new Date("2021-11-11"),
  price: 82000
}
})
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69a86a12aafccfc979374e93'),
    '1': ObjectId('69a86a12aafccfc979374e94'),
    '2': ObjectId('69a86a12aafccfc979374e95'),
    '3': ObjectId('69a86a12aafccfc979374e96'),
    '4': ObjectId('69a86a12aafccfc979374e97')
  }
}
Enterprise vehicles>
```

5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.



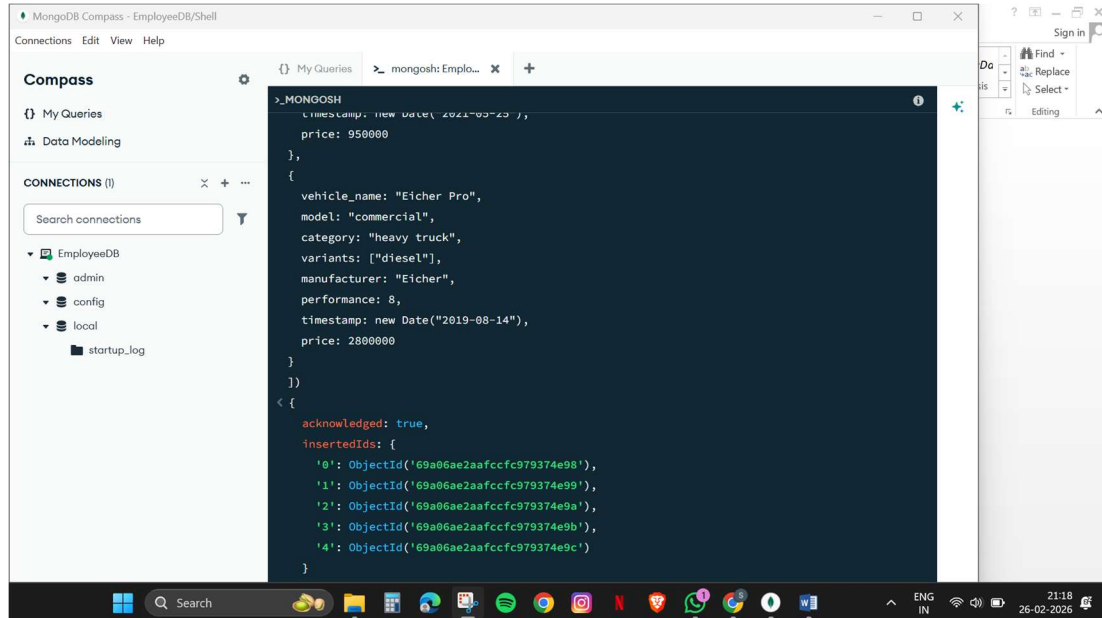
The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists 'EmployeeDB' with sub-items 'admin', 'config', 'local', and 'startup_log'. The main window displays the 'My Queries' tab with a MongoDB shell session. The session shows a successful insertMany operation into the 'four_wheelers' collection. The output includes the 'acknowledged' status (true) and a list of five 'insertedIds' (ObjectIds). The document structure for the first vehicle is visible: { 'vehicle_name': 'Maruti Swift', 'model': 'own', 'category': 'car', 'variants': ['vxi', 'zxi', 'petrol'], 'manufacturer': 'Maruti', 'performance': 8, 'timestamp': new Date('2022-02-02'), 'price': 750000 }.

```
> MONGOSH
> db.four_wheelers.insertMany([
  {
    vehicle_name: "Maruti Swift",
    model: "own",
    category: "car",
    variants: ["vxi", "zxi", "petrol"],
    manufacturer: "Maruti",
    performance: 8,
    timestamp: new Date("2022-02-02"),
    price: 750000
  },
  {
    vehicle_name: "Tata Ace",
    model: "commercial",
    category: "mini truck",
    variants: ["diesel"],
    manufacturer: "Tata",
    performance: 7,
    timestamp: new Date("2020-04-12"),
    price: 500000
  },
  {
    vehicle_name: "Ashok Leyland Bus",
```

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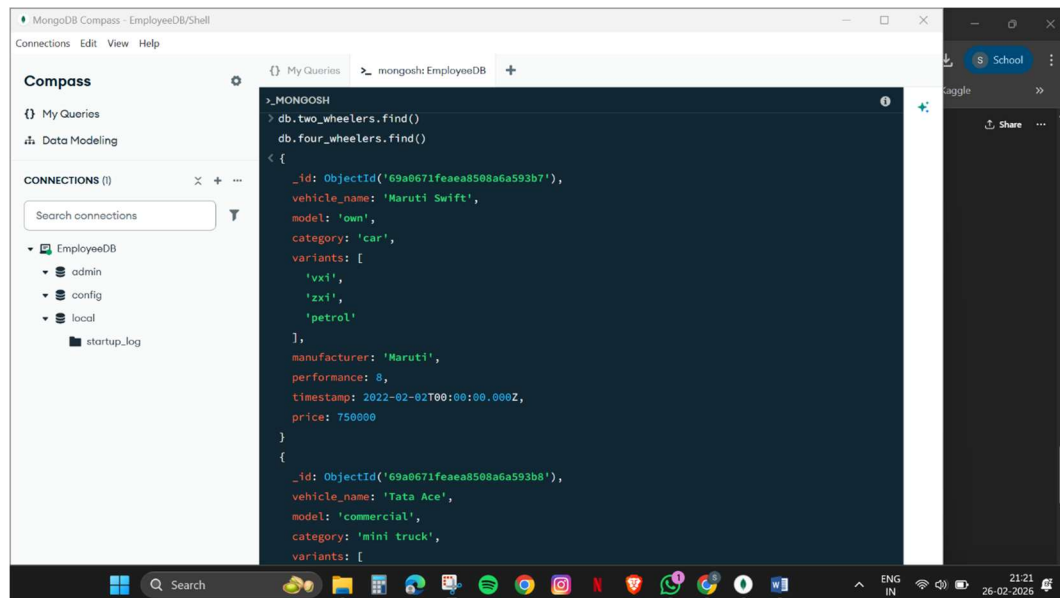
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The screenshot shows the MongoDB Compass interface with the 'EmployeeDB' selected. The 'My Queries' tab is active, and the shell displays the following command and its output:

```
>_MONGOSH
timesamp: new Date('2021-03-23'),
price: 950000
},
{
  vehicle_name: "Eicher Pro",
  model: "commercial",
  category: "heavy truck",
  variants: ["diesel"],
  manufacturer: "Eicher",
  performance: 8,
  timestamp: new Date("2019-08-14"),
  price: 2800000
}
})
<
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69a06ae2aafccfc979374e98'),
    '1': ObjectId('69a06ae2aafccfc979374e99'),
    '2': ObjectId('69a06ae2aafccfc979374e9a'),
    '3': ObjectId('69a06ae2aafccfc979374e9b'),
    '4': ObjectId('69a06ae2aafccfc979374e9c')
  }
}
```

6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.



The screenshot shows the MongoDB Compass interface with the 'EmployeeDB' selected. The 'My Queries' tab is active, and the shell displays the following command and its output:

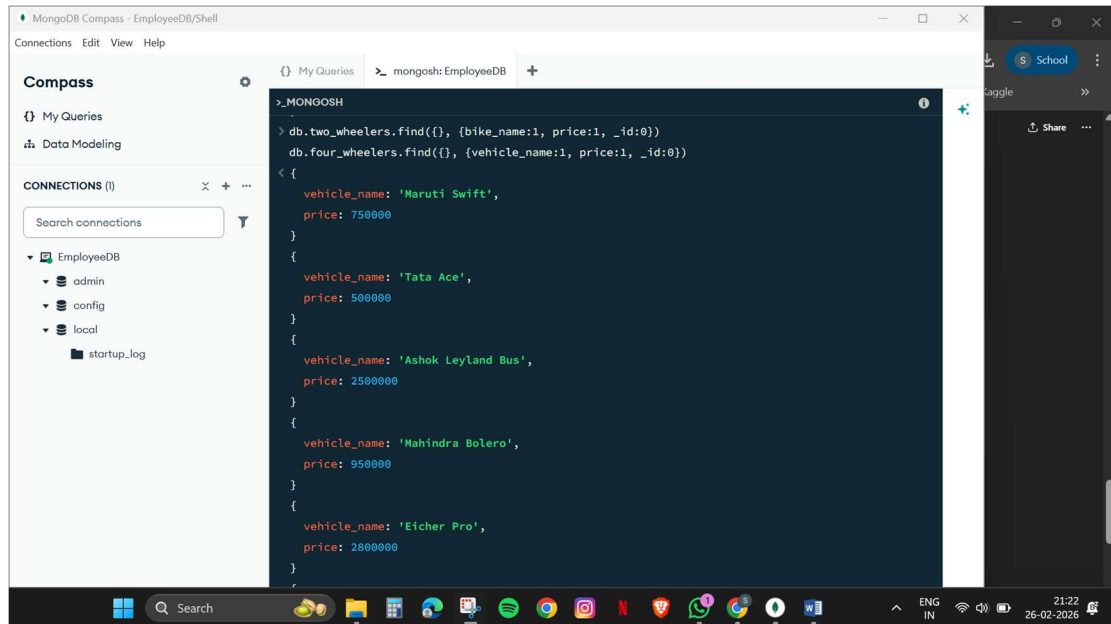
```
>_MONGOSH
db.two_wheelers.find()
db.four_wheelers.find()
<
{
  _id: ObjectId('69a0671feaea8508a6a593b7'),
  vehicle_name: 'Maruti Swift',
  model: 'own',
  category: 'car',
  variants: [
    'vxi',
    'zxi',
    'petrol'
  ],
  manufacturer: 'Maruti',
  performance: 8,
  timestamp: 2022-02-02T00:00:00.000Z,
  price: 750000
}
{
  _id: ObjectId('69a0671feaea8508a6a593b8'),
  vehicle_name: 'Tata Ace',
  model: 'commercial',
  category: 'mini truck',
  variants: [
```

7. Write a MongoDB query to display only vehicle name and price in all the collection of the database

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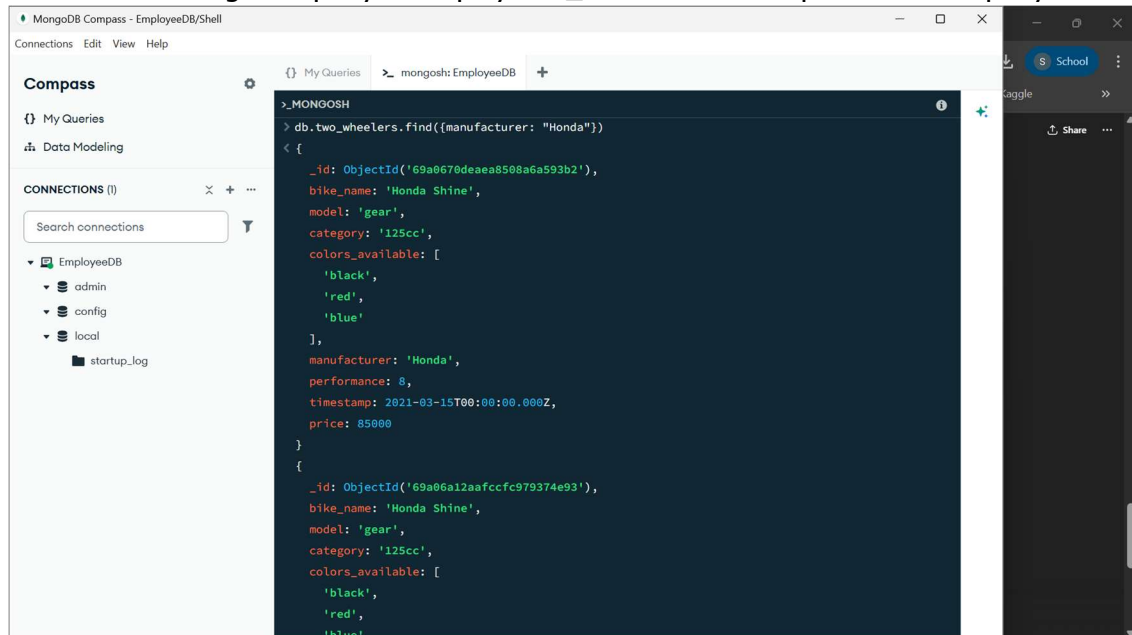
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The screenshot shows the MongoDB Compass interface. On the left, the 'Connections' panel lists 'EmployeeDB' with sub-items 'admin', 'config', 'local', and 'startup_log'. The main area displays the 'My Queries' tab for 'mongosh: EmployeeDB'. The shell prompt shows the following commands and results:

```
>_MONGOSH
> db.two_wheelers.find({}, {bike_name:1, price:1, _id:0})
db.four_wheelers.find({}, {vehicle_name:1, price:1, _id:0})
{
  vehicle_name: 'Maruti Swift',
  price: 750000
}
{
  vehicle_name: 'Tata Ace',
  price: 500000
}
{
  vehicle_name: 'Ashok Leyland Bus',
  price: 2500000
}
{
  vehicle_name: 'Mahindra Bolero',
  price: 950000
}
{
  vehicle_name: 'Eicher Pro',
  price: 2800000
}
```

8. Write a MongoDB query to display two_wheelers from a particular company



The screenshot shows the MongoDB Compass interface. On the left, the 'Connections' panel lists 'EmployeeDB' with sub-items 'admin', 'config', 'local', and 'startup_log'. The main area displays the 'My Queries' tab for 'mongosh: EmployeeDB'. The shell prompt shows the following command and results:

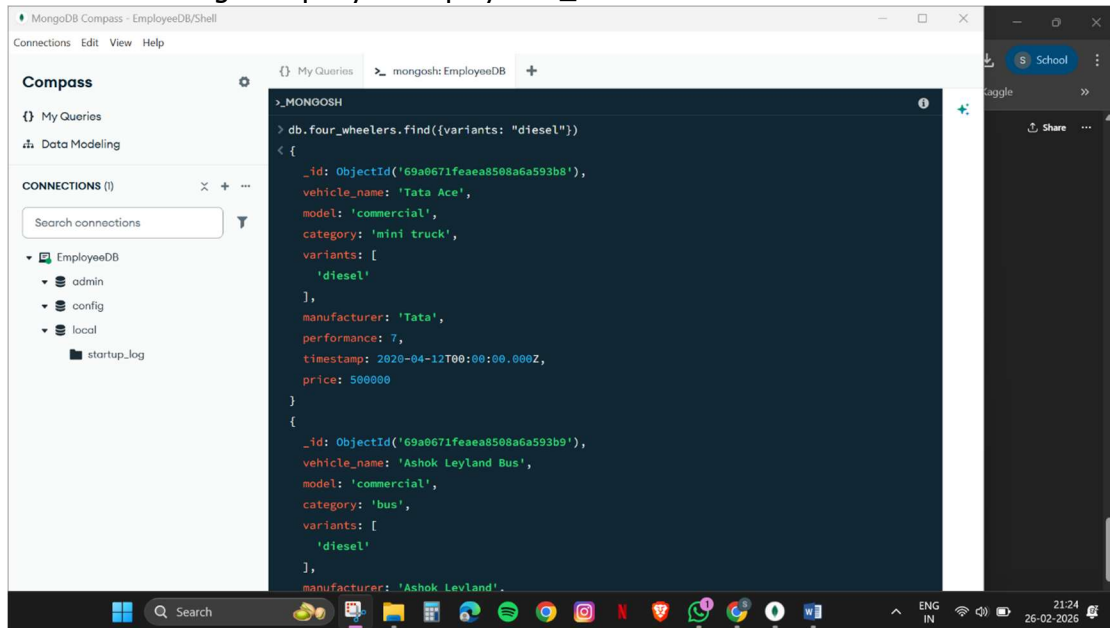
```
>_MONGOSH
> db.two_wheelers.find({manufacturer: "Honda"})
{
  _id: ObjectId('69a0670deaea8508a6a593b2'),
  bike_name: 'Honda Shine',
  model: 'gear',
  category: '125cc',
  colors_available: [
    'black',
    'red',
    'blue'
  ],
  manufacturer: 'Honda',
  performance: 8,
  timestamp: 2021-03-15T00:00:00.000Z,
  price: 85000
}
{
  _id: ObjectId('69a06a12aafccfc979374e93'),
  bike_name: 'Honda Shine',
  model: 'gear',
  category: '125cc',
  colors_available: [
    'black',
    'red',
    'blue'
  ],
  manufacturer: 'Honda',
  performance: 8,
  timestamp: 2021-03-15T00:00:00.000Z,
  price: 85000
}
```

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9. Write a MongoDB query to display four_wheelers available in diesel variants



The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'Connections' panel with 'EmployeeDB' selected. The main window shows the 'My Queries' tab with a query entered in the 'MONGOSH' shell:

```
> db.four_wheelers.find({variants: "diesel"})
```

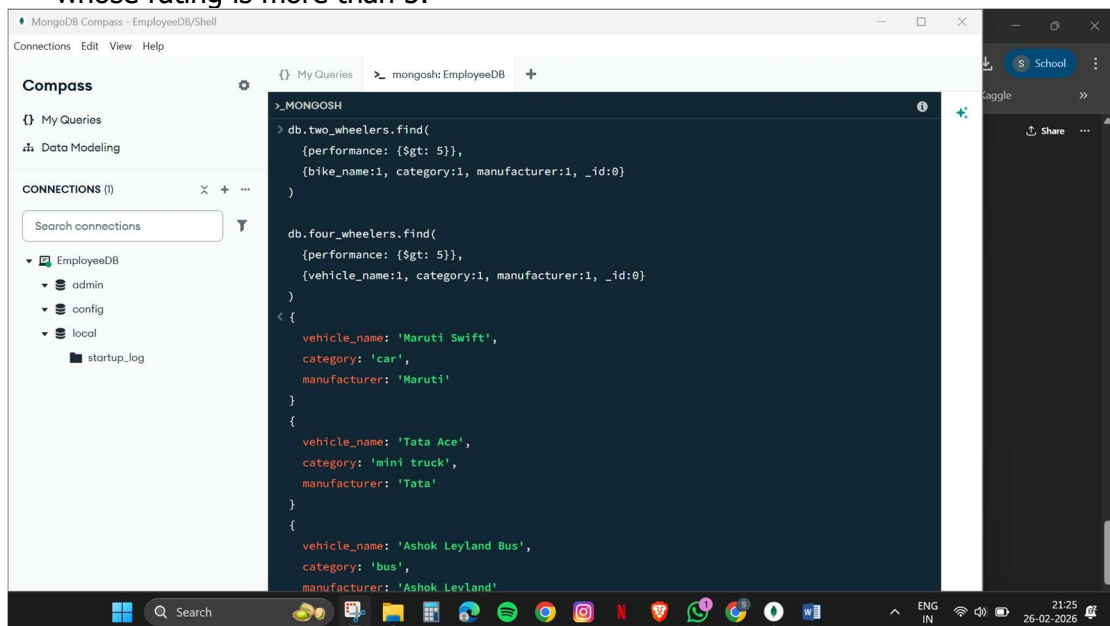
The query results are displayed in a JSON format:

```
{
  "_id": ObjectId("69a0671f6eaa508a6a593b8"),
  "vehicle_name": "Tata Ace",
  "model": "commercial",
  "category": "mini truck",
  "variants": [
    "diesel"
  ],
  "manufacturer": "Tata",
  "performance": 7,
  "timestamp": 2020-04-12T00:00:00.000Z,
  "price": 500000
}
```

The second result is partially visible:

```
{
  "_id": ObjectId("69a0671f6eaa508a6a593b9"),
  "vehicle_name": "Ashok Leyland Bus",
  "model": "commercial",
  "category": "bus",
  "variants": [
    "diesel"
  ],
  "manufacturer": "Ashok Leyland"
}
```

10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.



The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'Connections' panel with 'EmployeeDB' selected. The main window shows the 'My Queries' tab with a query entered in the 'MONGOSH' shell:

```
> db.two_wheelers.find(
  {performance: {$gt: 5}},
  {bike_name:1, category:1, manufacturer:1, _id:0}
)
```

The query results are displayed in a JSON format:

```
{
  "vehicle_name": "Maruti Swift",
  "category": "car",
  "manufacturer": "Maruti"
}
```

The second result is partially visible:

```
{
  "vehicle_name": "Tata Ace",
  "category": "mini truck",
  "manufacturer": "Tata"
}
```

The third result is partially visible:

```
{
  "vehicle_name": "Ashok Leyland Bus",
  "category": "bus",
  "manufacturer": "Ashok Leyland"
}
```

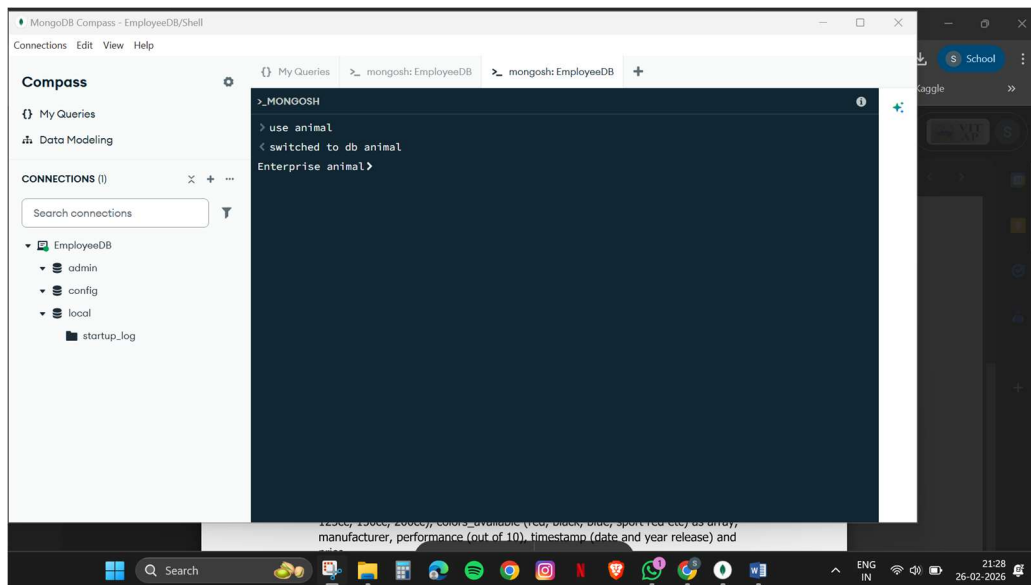
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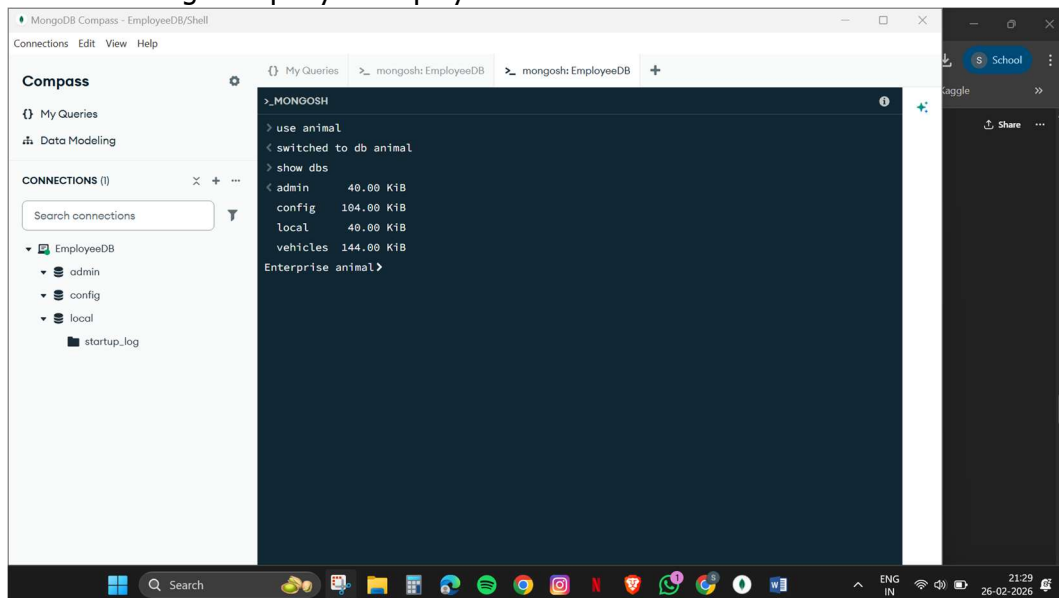
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2. Use MongoDB to implement the following DB operations for a Zoo

1. Create a database called 'animal' and *write* a MongoDB query to select database as 'animal'.



2. Write a MongoDB query to display all the databases.

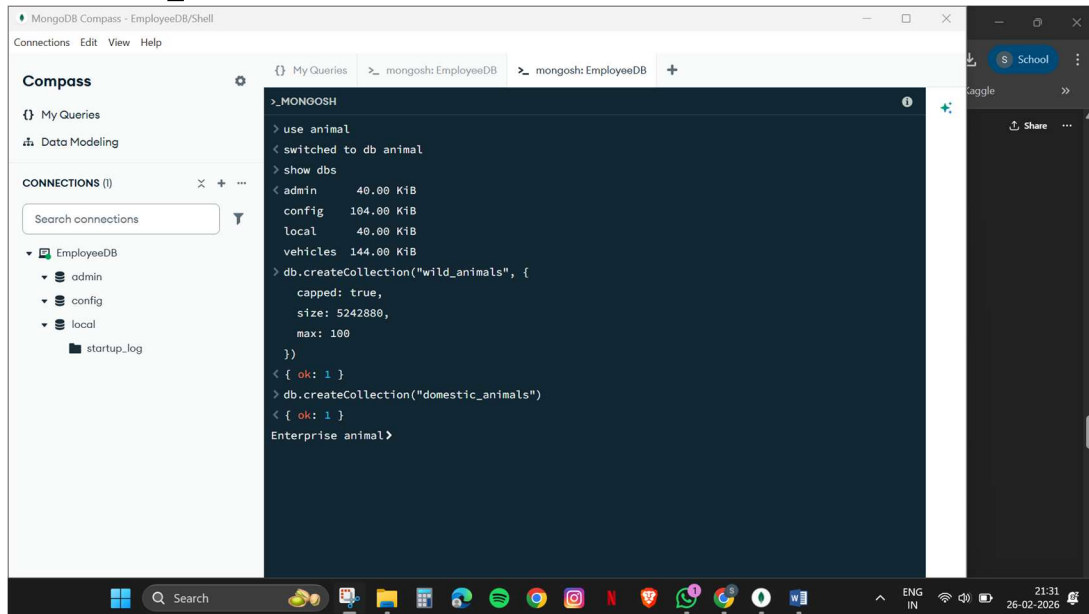


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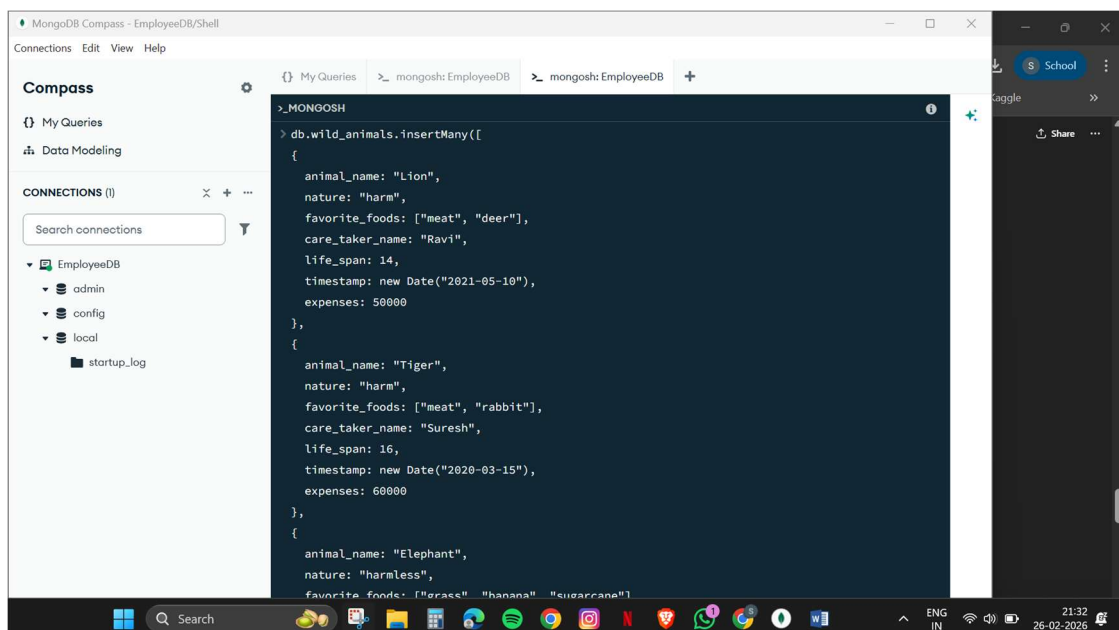
3. Create a collection called 'wild_animals'.(use capping) and Create a collection called 'domestic_animals'.



The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'Connections' list with 'EmployeeDB' selected. The main window shows the MongoDB Shell with the following commands and output:

```
> use animal
switched to db animal
> show dbs
admin 40.00 KiB
config 104.00 KiB
local 40.00 KiB
vehicles 144.00 KiB
> db.createCollection("wild_animals", {
  capped: true,
  size: 5242880,
  max: 100
})
{ ok: 1 }
> db.createCollection("domestic_animals")
{ ok: 1 }
Enterprise animal
```

4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.



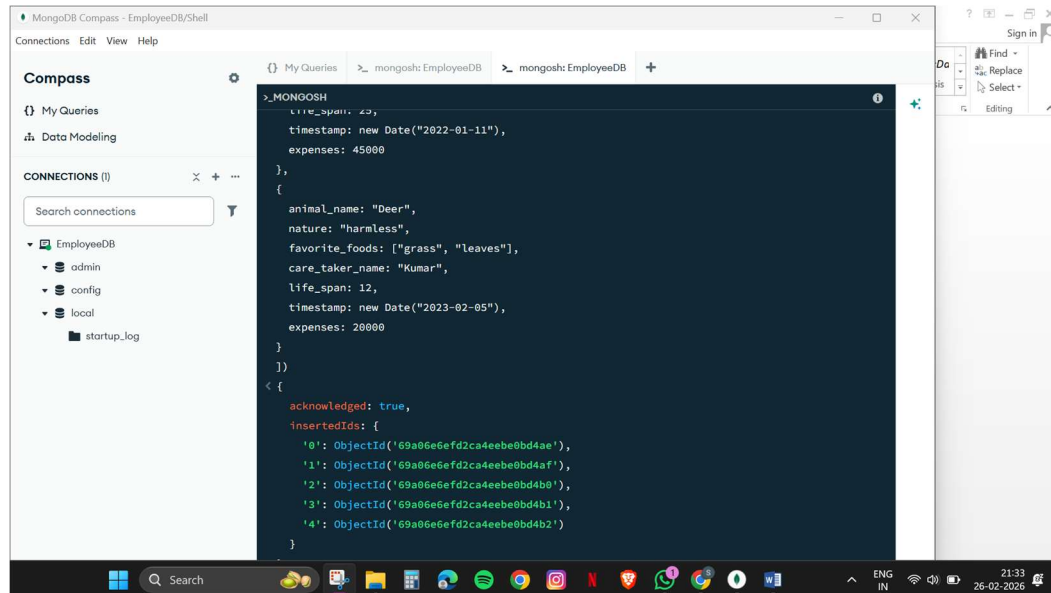
The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'Connections' list with 'EmployeeDB' selected. The main window shows the MongoDB Shell with the following command and output:

```
> db.wild_animals.insertMany([
  {
    animal_name: "Lion",
    nature: "harm",
    favorite_foods: ["meat", "deer"],
    care_taker_name: "Ravi",
    life_span: 14,
    timestamp: new Date("2021-05-10"),
    expenses: 50000
  },
  {
    animal_name: "Tiger",
    nature: "harm",
    favorite_foods: ["meat", "rabbit"],
    care_taker_name: "Suresh",
    life_span: 16,
    timestamp: new Date("2020-03-15"),
    expenses: 60000
  },
  {
    animal_name: "Elephant",
    nature: "harmless",
    favorite_foods: ["grass", "banana", "sugarcane"]
  }
])
```


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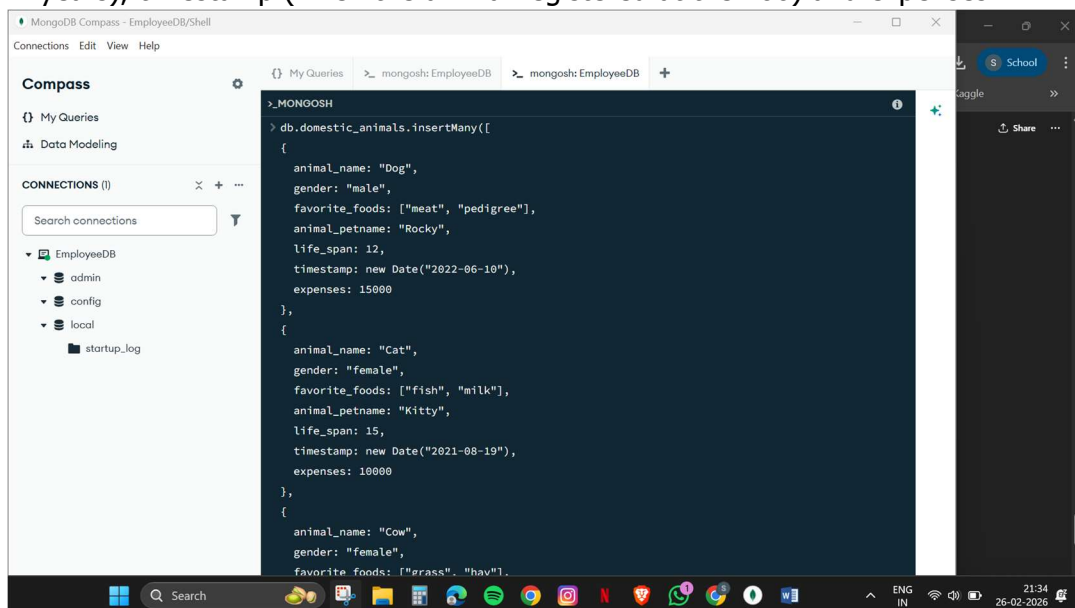
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The screenshot shows the MongoDB Compass interface with the mongoosh shell open. The shell displays the output of an insertMany command, showing five inserted documents with their respective IDs and timestamps.

```
> MONGOSH
> use EmployeeDB
> use collection
> insertMany(
  {
    timestamp: new Date("2022-01-11"),
    expenses: 45000
  },
  {
    animal_name: "Deer",
    nature: "harmless",
    favorite_foods: ["grass", "leaves"],
    care_taker_name: "Kumar",
    life_span: 12,
    timestamp: new Date("2023-02-05"),
    expenses: 20000
  }
)
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69a86e6fd2ca4eebe0bd4ae'),
    '1': ObjectId('69a86e6fd2ca4eebe0bd4af'),
    '2': ObjectId('69a86e6fd2ca4eebe0bd4b0'),
    '3': ObjectId('69a86e6fd2ca4eebe0bd4b1'),
    '4': ObjectId('69a86e6fd2ca4eebe0bd4b2')
  }
}
```

5. Add 5 domestic-animal details to the collection named 'domestic_animals'. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.



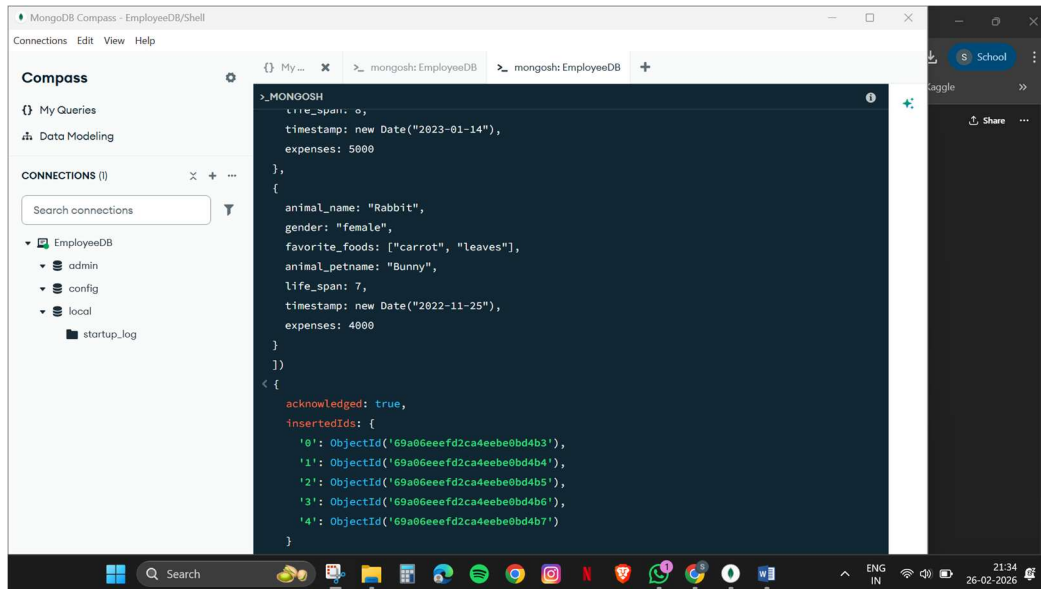
The screenshot shows the MongoDB Compass interface with the mongoosh shell open. The shell displays the command to insert five documents into the 'domestic_animals' collection.

```
> MONGOSH
> use EmployeeDB
> use collection
> insertMany(
  {
    animal_name: "Dog",
    gender: "male",
    favorite_foods: ["meat", "pedigree"],
    animal_petname: "Rocky",
    life_span: 12,
    timestamp: new Date("2022-06-10"),
    expenses: 15000
  },
  {
    animal_name: "Cat",
    gender: "female",
    favorite_foods: ["fish", "milk"],
    animal_petname: "Kitty",
    life_span: 15,
    timestamp: new Date("2021-08-19"),
    expenses: 10000
  },
  {
    animal_name: "Cow",
    gender: "female",
    favorite_foods: ["grass", "hay"]
  }
)
```

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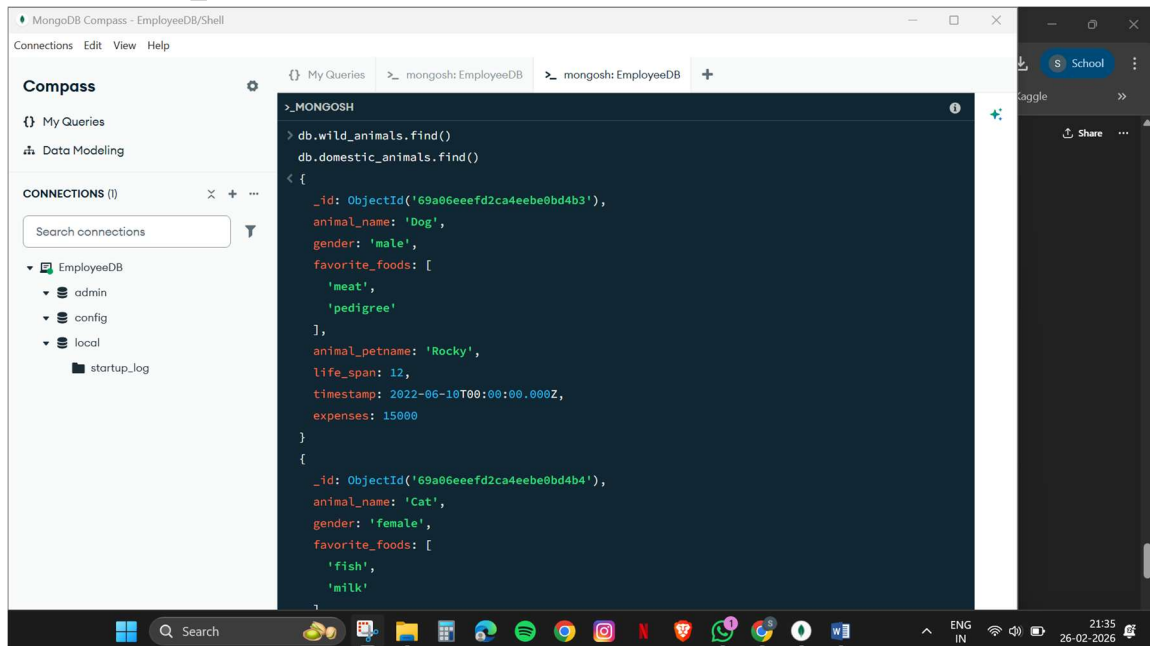
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The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'Connections' list with 'EmployeeDB' selected. The main window shows a query result for a collection named 'wild_animals'. The query is `> db.wild_animals.find()`. The result is a JSON array of documents. The first document is:

```
{
  "animal_name": "Rabbit",
  "gender": "female",
  "favorite_foods": ["carrot", "leaves"],
  "animal_petname": "Bunny",
  "life_span": 7,
  "timestamp": new Date("2022-11-25"),
  "expenses": 4000
}
```

6. Write a MongoDB query to display all documents available in `wild_animals` and `domestic_animals`.



The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'Connections' list with 'EmployeeDB' selected. The main window shows a query result for a collection named 'wild_animals'. The query is `> db.wild_animals.find()`. The result is a JSON array of documents. The first document is:

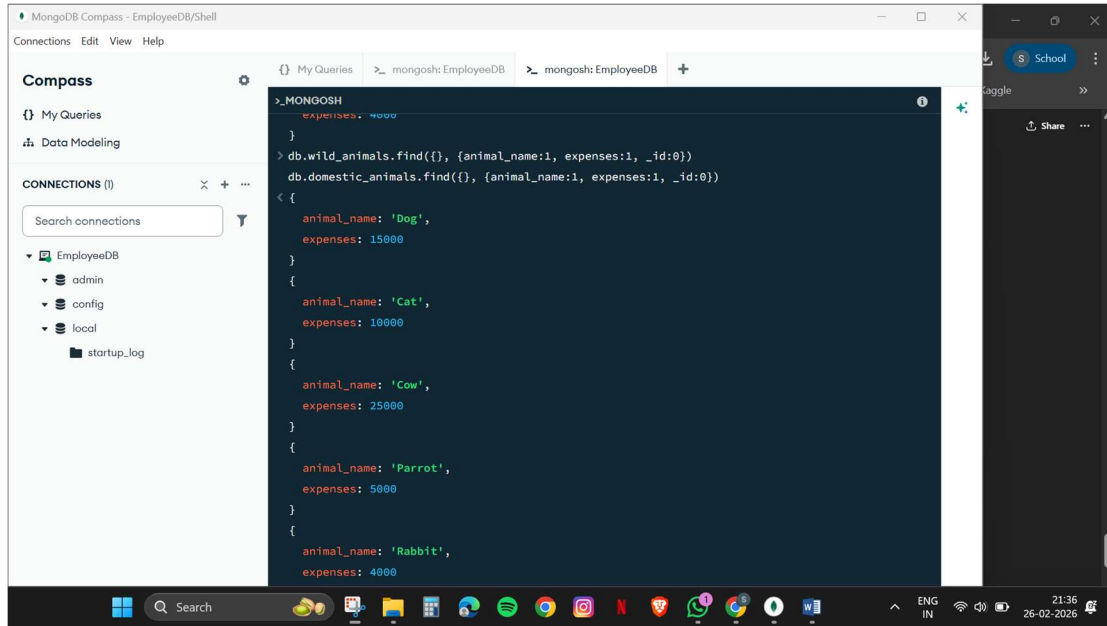
```
{
  "_id": ObjectId('69a06eeefd2ca4eebe8bd4b3'),
  "animal_name": 'Dog',
  "gender": 'male',
  "favorite_foods": [
    'meat',
    'pedigree'
  ],
  "animal_petname": 'Rocky',
  "life_span": 12,
  "timestamp": 2022-06-10T00:00:00.000Z,
  "expenses": 15000
}
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech
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Date: 26.02.2026
School: SCOPE
Reg. no.: 23BCE8922

7. Write a MongoDB query to display only animal name and expenses in all the collection of the database

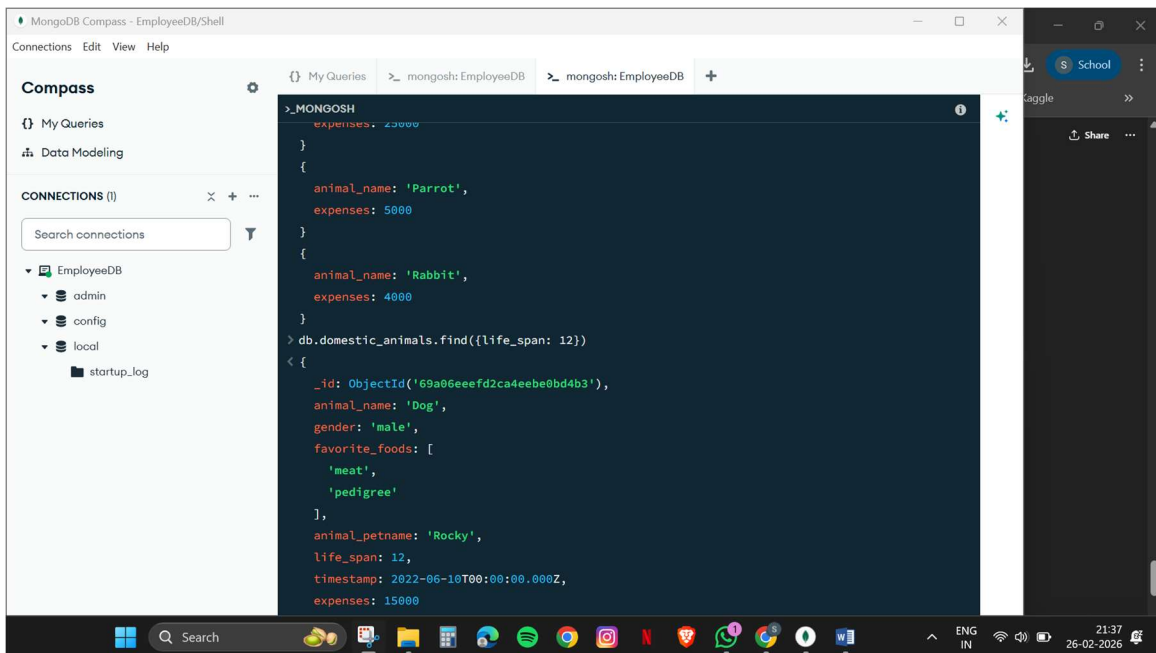


The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'EmployeeDB' database with collections 'admin', 'config', and 'local'. The main window shows the MongoDB Shell with the following commands and results:

```
> use EmployeeDB
> db.wild_animals.find({}, {animal_name:1, expenses:1, _id:0})
db.domestic_animals.find({}, {animal_name:1, expenses:1, _id:0})
```

```
{
  animal_name: 'Dog',
  expenses: 15000
}
{
  animal_name: 'Cat',
  expenses: 10000
}
{
  animal_name: 'Cow',
  expenses: 25000
}
{
  animal_name: 'Parrot',
  expenses: 5000
}
{
  animal_name: 'Rabbit',
  expenses: 4000
}
```

8. Write a MongoDB query to display domestic_animals whose life is a particular year



The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'EmployeeDB' database with collections 'admin', 'config', and 'local'. The main window shows the MongoDB Shell with the following commands and results:

```
> use EmployeeDB
> db.domestic_animals.find({life_span: 12})
```

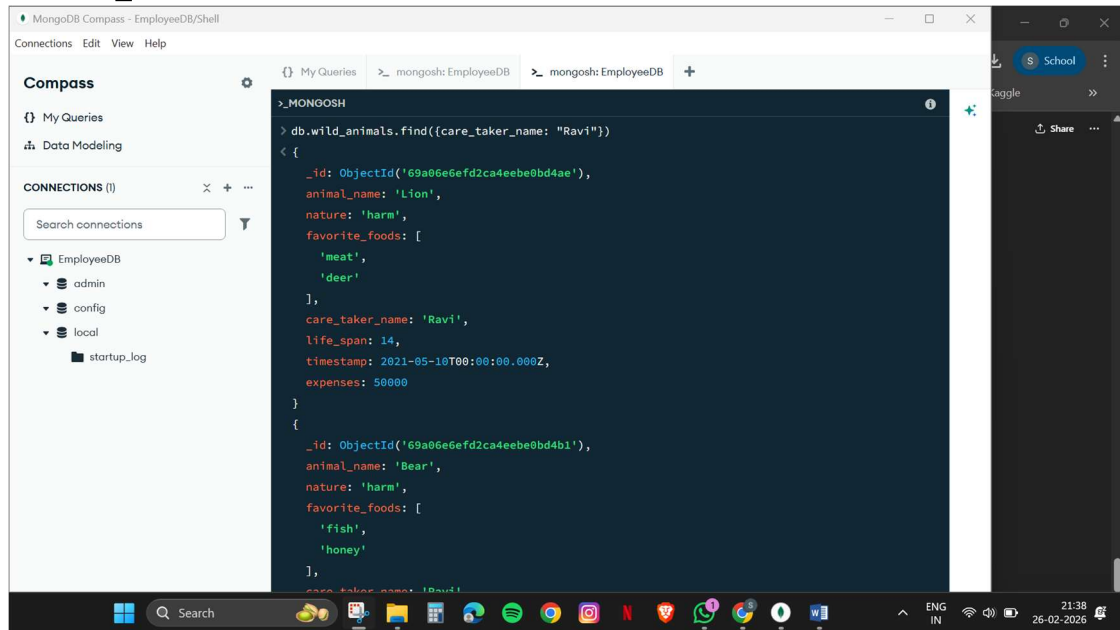
```
{
  _id: ObjectId('69a06eeefd2ca4eebe0bd4b3'),
  animal_name: 'Dog',
  gender: 'male',
  favorite_foods: [
    'meat',
    'pedigree'
  ],
  animal_petname: 'Rocky',
  life_span: 12,
  timestamp: 2022-06-10T00:00:00.000Z,
  expenses: 15000
}
```

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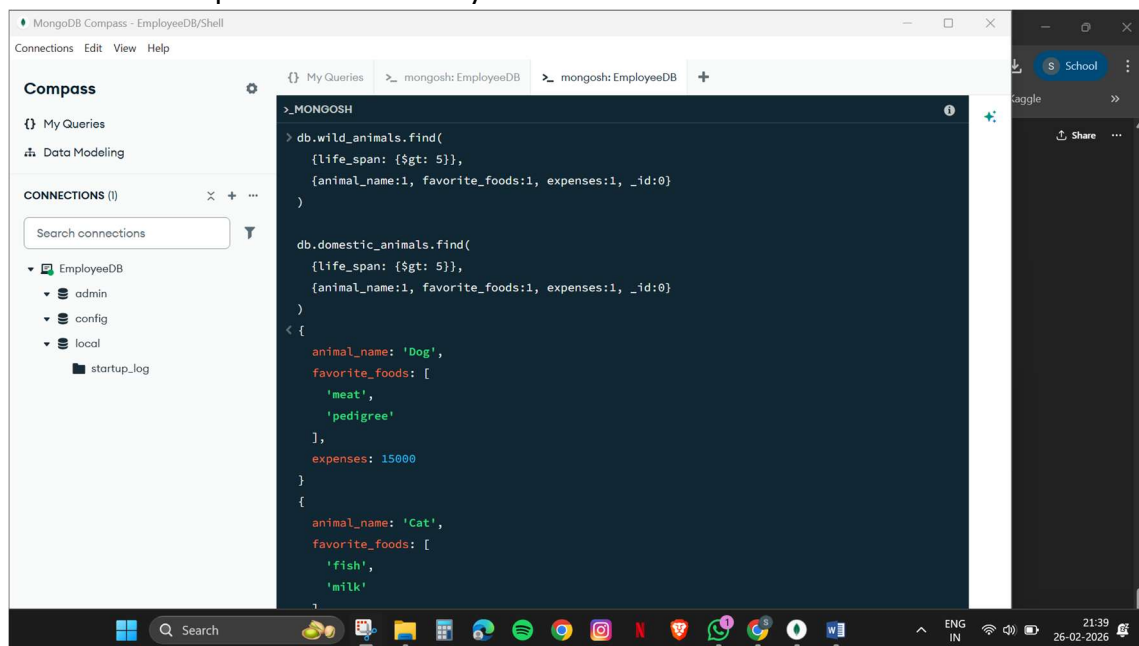
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9. Write a MongoDB query to display wild_animals available under a particular care_taker



The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'EmployeeDB' database with collections 'admin', 'config', and 'local'. The main panel shows a query in the 'My Queries' tab: `db.wild_animals.find({care_taker_name: "Ravi"})`. The results are displayed in a JSON format, showing two documents for 'Ravi': one for a 'Lion' and one for a 'Bear'. The 'Lion' document includes fields for '_id', 'animal_name', 'nature', 'favorite_foods', 'care_taker_name', 'life_span', 'timestamp', and 'expenses'. The 'Bear' document includes fields for '_id', 'animal_name', 'nature', 'favorite_foods', and 'care_taker_name'.

10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.



The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'EmployeeDB' database with collections 'admin', 'config', and 'local'. The main panel shows a query in the 'My Queries' tab: `db.wild_animals.find({life_span: {$gt: 5}})`. The results are displayed in a JSON format, showing two documents for 'Ravi': one for a 'Lion' and one for a 'Bear'. The 'Lion' document includes fields for '_id', 'animal_name', 'nature', 'favorite_foods', 'care_taker_name', 'life_span', 'timestamp', and 'expenses'. The 'Bear' document includes fields for '_id', 'animal_name', 'nature', 'favorite_foods', and 'care_taker_name'.